

Project Name	Import Data using Transform Maps	Maximum Marks	4 Marks
Date	31 January 2025	Team ID	6ab615fddcdcae33acb38af6f

Technology Stack

Technical architecture: The data import and governance pipeline processes legacy CSV training records through staging tables, runs Transform Map coalesce logic, resolves manager references by dot-walking, and enforces role-based ACL security.

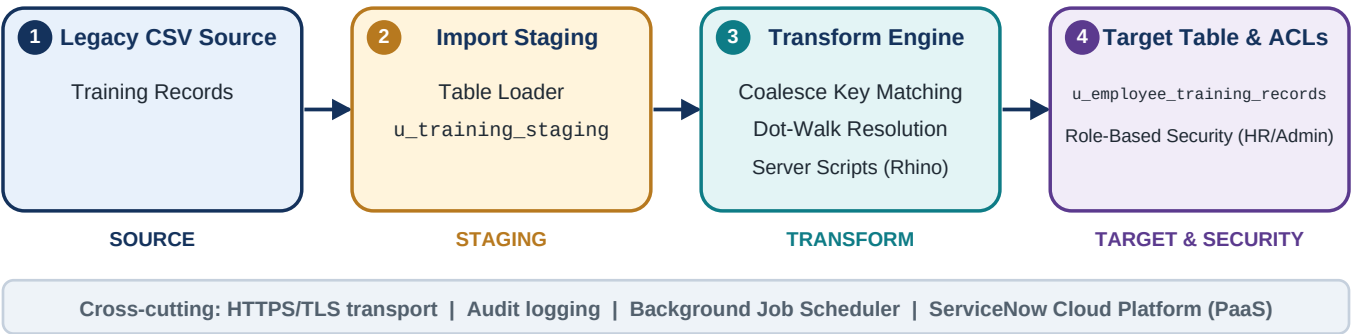


Table-1: Components & Technologies

S.No	Component	Description	Technology	Pipeline Stage
1	User Interface	ServiceNow workspace and admin forms for managing import sets and Transform Maps.	HTML, CSS, JavaScript, Jelly UI	Source
2	Application Logic-1	Field mapping rules and Transform Map script execution for data transformation.	Server-side JavaScript (Rhino)	Transform
3	Application Logic-2	Reference resolution and parent manager lookup through dot-walking.	GlideRecord API / Dot-Walk Engine	Transform
4	Application Logic-3	Security enforcement that restricts confidential training scores.	Access Control Lists (ACLs) / Roles	Target & Security
5	Database	Relational schema for employee training records and lookup reference fields.	ServiceNow CMDb / Custom Tables	Target & Security
6	Cloud Database	Enterprise multi-tenant database backend hosting instance data.	ServiceNow Enterprise Cloud DB	Platform
7	File Storage	Storage for uploaded legacy training CSV files and attachments.	ServiceNow Attachment DB / BLOB	Source
8	External API-1	Inbound REST API endpoints for automated legacy data feed ingestion.	ServiceNow REST Table API	Source
9	External API-2	Active Directory / LDAP integration for employee identity verification.	LDAP Integration Directory	Security
10	Machine Learning Model	Automated duplicate record detection and data quality heuristics.	ServiceNow Predictive Intelligence	Transform
11	Infrastructure (Server / Cloud)	Enterprise PaaS deployment that hosts the application instance securely.	ServiceNow Cloud Platform (PaaS)	Platform

Table-2: Application Characteristics

S.No	Characteristic	Description	Technology
1	Open-Source Frameworks	Open-source libraries used within the ServiceNow development framework.	AngularJS, Prototype.js, Rhino JS
2	Security Implementations	Role-based access control, table-level and field-level ACLs, and encrypted transport.	RBAC, ACLs, HTTPS/TLS
3	Scalable Architecture	Multi-tenant SaaS cloud architecture with asynchronous worker queues.	ServiceNow Job Scheduler / Node Clustering
4	Availability	High-availability cloud infrastructure with automated failover and redundancy.	Distributed Cloud Server Clustering
5	Performance	Optimized transform batch processing and database indexing on coalesce fields.	Bulk Import Sets, DB Indexing, Caching

Stack Summary by Pipeline Stage

Stage	Main Technologies	Purpose
Source	User Interface, File Storage, REST Table API	Receive legacy CSV files and automated feeds.
Staging	Import Set loader (u_training_staging)	Hold raw rows without touching production data.
Transform	Rhino server scripts, GlideRecord / dot-walking, Predictive Intelligence	Match, map, resolve references, and check data quality.
Target & Security	Custom tables, ACLs and roles, LDAP	Store validated records and restrict access to scores.
Platform	Enterprise Cloud DB, ServiceNow PaaS	Provide multi-tenant hosting, availability, and scale.

References

- ServiceNow Documentation: Import Sets and Transform Maps Guide (<https://docs.servicenow.com>)
- ServiceNow Developer Portal: Scripting and GlideRecord API Reference (<https://developer.servicenow.com>)
- ServiceNow Access Control Lists (ACL) Best Practices and Security Hardening
- IBM Cloud Architecture & C4 Model Standards (<https://c4model.com> / <https://www.ibm.com/cloud/architecture>)